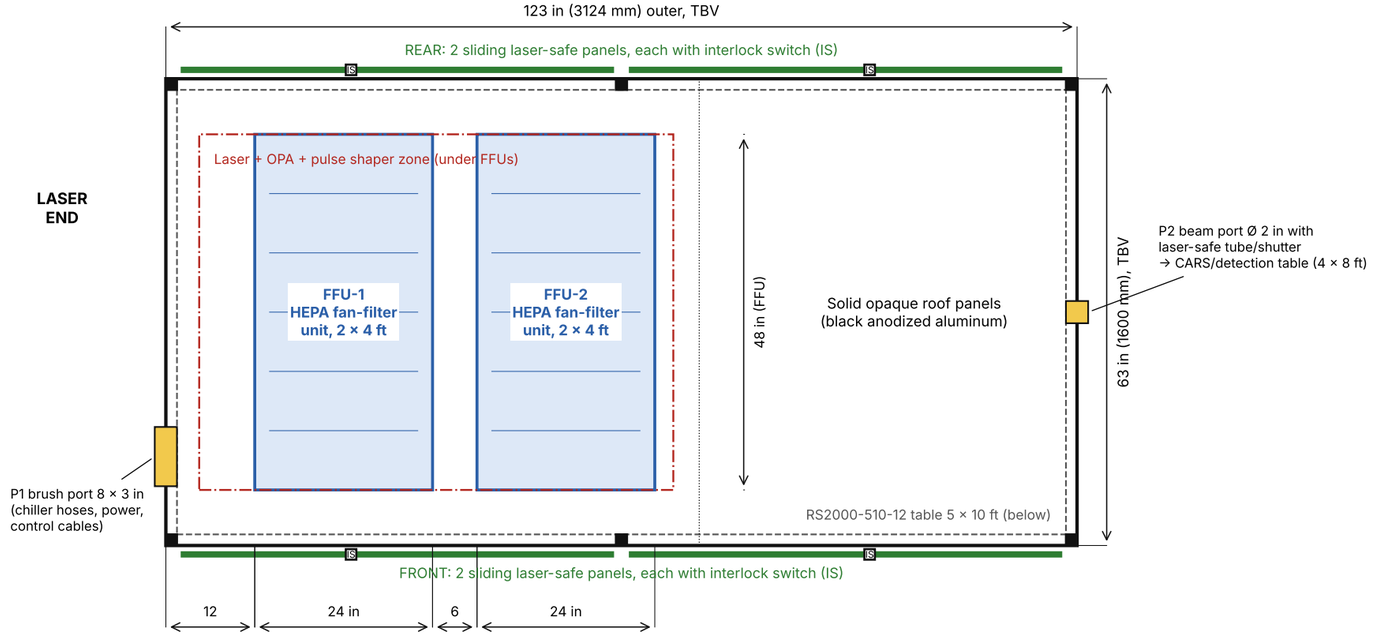
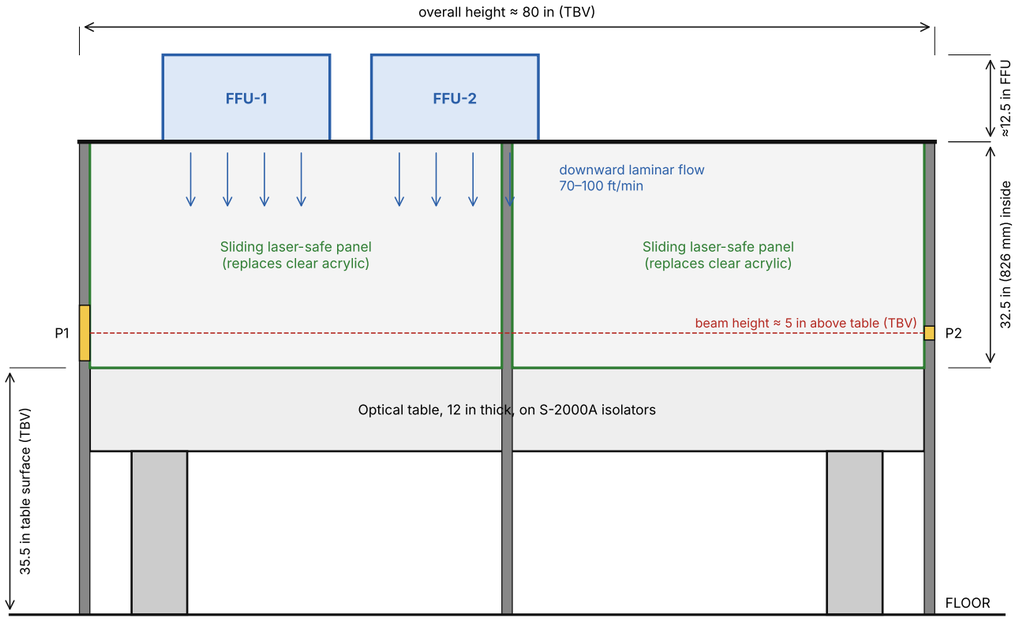
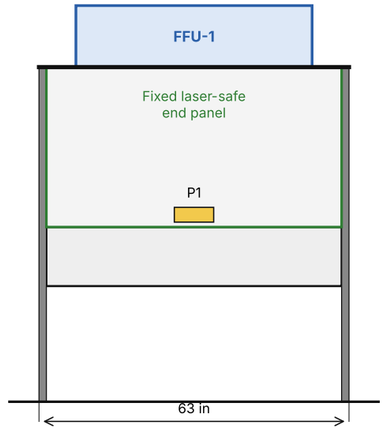
PLAN VIEW (roof shown)



FRONT ELEVATION (long side)



END ELEVATION (laser end)



DESIGN INTENT

Concept drawing for a BUDGETARY quotation of options on Newport TE80-510 (Quote Q-296033). Not for fabrication; Newport to confirm all dimensions and details. TBV = to be verified.

1. BASE ENCLOSURE

TE80-510 for RS2000-510-12 (5 × 10 ft × 12 in) table: floor-standing 80/20 frame, 32.5 in (826 mm) panel height, sliding panels. Frame must not touch the table, so FFU fan vibration does not reach the optics.

2. HEPA FAN-FILTER UNITS (FFU-1, FFU-2)

Qty 2, nominal 2 × 4 ft, roof-mounted over the laser/OPA/shaper zone. HEPA ≥ 99.99 % at 0.3 µm (H14 or equal); EC motor with speed control; face velocity 70–100 ft/min; sound ≤ 55 dBA at 1 m; 120 V, 60 Hz. Isolation gaskets between FFU and roof frame. Pre-filter accessible from above.

3. ROOF

Remaining roof area closed with opaque black-anodized aluminum panels (laser-safe). Slight positive pressure; air exits under the panel skirt.

4. LASER-SAFE PANELS (replace clear acrylic)

All side, end and sliding panels laser-safe for: (a) Yb amplifier 1030 nm, 40 W average, 400 µJ, < 250 fs, 100 kHz; (b) OPA 650–900 nm, a few W, < 55 fs; idler 1200–2500 nm. Preferred: opaque black-anodized aluminum. If viewing windows are offered, OD ≥ 7 at 1030 nm and ≥ 5 at 650–900 nm. Vendor to state the direct-exposure rating (e.g. per IEC 60825-4).

5. INTERLOCKS

Each sliding panel (4 total) fitted with an interlock switch (IS), normally-open contacts, wired to a terminal block for the laser interlock circuit.

6. PORTS

P1: laser-end brush/baffled port ≈ 8 × 3 in at table level for chiller hoses, power and control cables. P2: Ø 2 in beam port on the far end at beam height, with laser-safe tube and removable cap, toward the 4 × 8 ft CARS table.

7. QUOTE REQUEST

Please price as separate lines: FFU-1/FFU-2 with roof frame; opaque roof panels; laser-safe panel upgrade; 4 interlock switches; ports P1 and P2. Include lead time and installation if offered.

HEPA FFU · laser-safe panel · port · laser zone / beam

TE80-510 ENCLOSURE: HEPA FAN-FILTER UNITS + LASER-SAFE PANELS (CONCEPT, DRAFT)	Drawn: M. Badr Date: 3 Nov 2026 Rev: A (for budgetary quote)
Morgan State University, Clarence M. Mitchell, Jr. School of Engineering FY2027 DURIP fs-CARS system, item 11 · PI: Dr. Oluwatobi Busari	Units: inches (mm) Scale: not to scale Sheet 1 of 1